

Static Analysis and Verification

SAV 2020/2021

Ondřej Lengál, Tomáš Vojnar

`{lengal,vojnar}@fit.vutbr.cz`

**Brno University of Technology
Faculty of Information Technology
Božetěchova 2, 612 66 Brno**

Abstract Interpretation

Introduction

- Compared to model checking in which the stress is put on a systematic execution of a system being verified (or its model), the emphasis in **static analysis** is on minimizing the amount of execution of the code.

It is either **not executed at all** (the case of looking for bug patterns) or just **on some abstract level**, typically with an in advance fixed abstraction (data flow analysis, **abstract interpretation**, ...).

- However, the borderline between model checking and static analysis is not sharp (especially when considering **abstract interpretation** and model checking based on predicate abstraction).
- Many static analyses are such that they can be applied to parts of code without the need to describe their environment.
- **Static analysis approaches**: bug pattern searching, type analysis, data flow analysis, ..., **abstract interpretation**, (and sometimes even model checking).

Static Analyses

- **Efficiency** (**effectiveness**) of an analysis often crucially depends on the **abstraction** used.
- An analysis **successful** for one class of programs/properties to be checked may (and is very likely to) **fail** for a different one: too much imprecision, inefficiency, divergence.
- Analyses are tailored for specific classes of programs and their properties of interest.
 - This implies a need to prove **soundness** (**completeness**) of each analysis.

Abstract Interpretation

- Introduced by Patrick and Radhia Cousot at POPL'77.
- A general **framework** for static analyses.
- Particular analyses are created by providing specific **components** (abstract domain, abstract transformers, ...) to the framework.
- Abstract interpretation assigns a program an **abstract semantics** over an **abstract domain** and analyses this abstract semantics (cf. predicate abstraction).
- When certain properties of the components are met (wrt the concrete semantics), the analysis is guaranteed to be **sound**.

Ingredients of Abstract Interpretation

- **Abstract domain**
 - a set of **abstract contexts**,
 - an abstract context represents a set of program states (typically used to represent a set of program states reachable at some program location).
- **Abstract transformers**
 - for each program operation there is a corresponding transformer that represents the effect of the operation performed on an abstract context.
- **Join operator**
 - combines abstract contexts from several branches into a single one.
- **Widening**
 - performed on a sequence of abstract contexts appearing at a given location to accelerate obtaining a **fixpoint**.
- **Narrowing**
 - may be used to refine the result of **widening**.

Abstract Interpretation — formally

- **Abstract interpretation** I of a program P with the instruction set Instr is a tuple

$$I = (Q, \sqcup, \sqsubseteq, \top, \perp, \tau)$$

where

- Q is the **abstract domain** (domain of **abstract contexts**),
 - $\sqcup : Q \times Q \rightarrow Q$ is the **join operator** for accumulation of abstract contexts,
 - $\sqsubseteq \subseteq Q \times Q$ is an ordering defined as $x \sqsubseteq y \iff x \sqcup y = y$ where $(Q, \sqsubseteq)$ is a **complete (join semi-)lattice**,
 - $\top \in Q$ is the supremum of $(Q, \sqsubseteq)$,
 - $\perp \in Q$ is the (single) infimum of $(Q, \sqsubseteq)$,
 - $\tau : \text{Instr} \times Q \rightarrow Q$ defines the **abstract transformers** for particular instructions, required to be **monotone** on Q for each instruction from Instr .
- The **soundness** of abstract interpretation may be guaranteed using **Galois connections**.

Galois Connections

- **Galois connection** is a quadruple $\pi = (\mathcal{P}, \alpha, \gamma, \mathcal{Q})$ such that:
 - $\mathcal{P} = \langle P, \leq \rangle$ and $\mathcal{Q} = \langle Q, \sqsubseteq \rangle$ are *partially ordered sets* (posets),
 - $\alpha : P \rightarrow Q$ and $\gamma : Q \rightarrow P$ are functions such that $\forall p \in P$ and $\forall q \in Q$:

$$p \leq \gamma(q) \iff \alpha(p) \sqsubseteq q$$

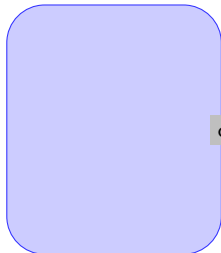
Galois Connections

- **Galois connection** is a quadruple $\pi = (\mathcal{P}, \alpha, \gamma, \mathcal{Q})$ such that:
 - $\mathcal{P} = \langle P, \leq \rangle$ and $\mathcal{Q} = \langle Q, \sqsubseteq \rangle$ are *partially ordered sets* (posets),
 - $\alpha : P \rightarrow Q$ and $\gamma : Q \rightarrow P$ are functions such that $\forall p \in P$ and $\forall q \in Q$:

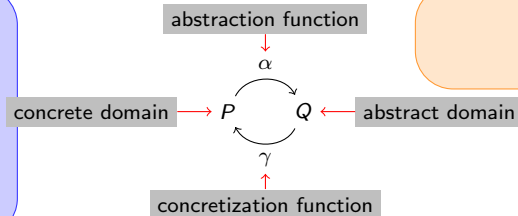
$$p \leq \gamma(q) \iff \alpha(p) \sqsubseteq q$$

- In abstract interpretation, Q is the *abstract domain* and P a (more) *concrete domain* – elements of both domains, called *abstract/concrete contexts*, represent sets of states:

concrete contexts such as:



abstract contexts such as:



Galois Connections

- **Galois connection** is a quadruple $\pi = (\mathcal{P}, \alpha, \gamma, \mathcal{Q})$ such that:
 - $\mathcal{P} = \langle P, \leq \rangle$ and $\mathcal{Q} = \langle Q, \sqsubseteq \rangle$ are *partially ordered sets* (posets),
 - $\alpha : P \rightarrow Q$ and $\gamma : Q \rightarrow P$ are functions such that $\forall p \in P$ and $\forall q \in Q$:

$$p \leq \gamma(q) \iff \alpha(p) \sqsubseteq q$$

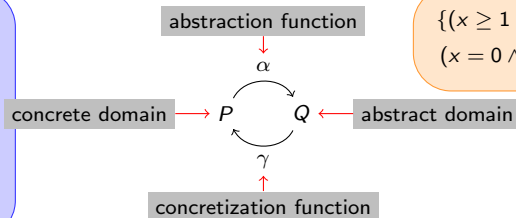
- In abstract interpretation, Q is the *abstract domain* and P a (more) *concrete domain* – elements of both domains, called *abstract/concrete contexts*, represent sets of states:

concrete contexts such as:

$\{\{x \mapsto 1, y \mapsto 0\},$
 $\{x \mapsto 2, y \mapsto 0\},$
 $\{x \mapsto 3, y \mapsto 0\},$
 $\{x \mapsto 0, y \mapsto 1\},$
 $\{x \mapsto 0, y \mapsto 2\},$
 $\{x \mapsto 0, y \mapsto 3\}\}$

abstract contexts such as:

$\{(x \geq 1 \wedge y = 0),$
 $(x = 0 \wedge y \geq 1)\}$



Galois Connections

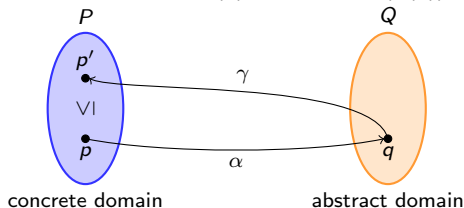
- **Implication:** if the abstraction and concretization functions of an abstract interpretation form a Galois connection, $\forall p \in P. p \leq \gamma(\alpha(p))$.

Proof.

Take any $p \in P$, and let $q = \alpha(p)$.

As $q = \alpha(p) \Rightarrow \alpha(p) \sqsubseteq q$, the definition of Galois connections implies $p \leq \gamma(q)$.

By plugging in the fact that $q = \alpha(p)$, we get $p \leq \gamma(\alpha(p))$.



□

Galois Connections

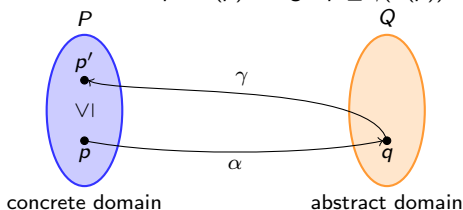
- **Implication:** if the abstraction and concretization functions of an abstract interpretation form a Galois connection, $\forall p \in P. p \leq \gamma(\alpha(p))$.

Proof.

Take any $p \in P$, and let $q = \alpha(p)$.

As $q = \alpha(p) \Rightarrow \alpha(p) \sqsubseteq q$, the definition of Galois connections implies $p \leq \gamma(q)$.

By plugging in the fact that $q = \alpha(p)$, we get $p \leq \gamma(\alpha(p))$.



- If each instruction i from `Instr` and the corresponding abstract transformer τ_i respect the Galois connection, i.e., for each $p \in P$,

$$\alpha(i(p)) \sqsubseteq \tau_i(\alpha(p)),$$

the abstract interpretation may only over-approximate the concrete semantics. Hence, it is **sound**.

Fixpoint Approximation

- The analysis of a program through an abstract interpretation may be viewed as finding the **least/greatest fixpoint** of the equation $\bar{A} = \bar{\tau}(\bar{A})$ where $\bar{A}$ is a vector of abstract contexts (one per program location) and $\bar{\tau}$ is an extension of τ to the entire program.
- By Knaster-Tarski, these fixpoints exist.
- In some cases (e.g., given a program with loops and using an infinite domain), computation of the *most precise* abstract fixpoint is **not generally guaranteed to terminate** (consider *id* as the abstraction function).
- To guarantee termination, the fixpoint can be approximated. This is done using the following two operations:
 - **widening**: performs an over-approximation of a fixpoint,
 - **narrowing**: refines an approximation of a fixpoint.
- Neither widening nor narrowing are necessary, but at least widening is often convenient. Narrowing may be sometimes missing (e.g., in polyhedral analysis).

Widening

- Let $I = (Q, \sqcup, \sqsubseteq, \top, \perp, \tau)$ be an abstract interpretation of a program.
- The binary **widening** operation ∇ is defined as:
 - $\nabla : Q \times Q \rightarrow Q$,
 - $\forall C, D \in Q : (C \sqcup D) \sqsubseteq (C \nabla D)$,
 - for all (non-strictly) increasing infinite sequences $C_0 \sqsubseteq C_1 \sqsubseteq \dots \sqsubseteq C_n \sqsubseteq \dots$, it holds that the infinite sequence $s_0, s_1, \dots, s_n, \dots$ defined recursively as

$$\begin{aligned}s_0 &= C_0, \\ s_n &= s_{n-1} \nabla C_n\end{aligned}$$

is not strictly increasing (and because the result of ∇ is an upper bound, the sequence eventually stabilizes).

- Widening can be applied later in the computation, the later it is applied the more precise is the result (but the computation takes longer time).

Narrowing

- Let $I = (Q, \sqcup, \sqsubseteq, \top, \perp, \tau)$ be an abstract interpretation of a program.
- The binary **narrowing** operation Δ is defined as:
 - $\Delta: Q \times Q \rightarrow Q$,
 - $\forall C, D \in Q : C \sqsubseteq D \Rightarrow (C \sqsubseteq (C \Delta D) \sqsubseteq D)$,
 - for all (non-strictly) decreasing infinite sequences $C_0 \sqsupseteq C_1 \sqsupseteq \dots \sqsupseteq C_n \sqsupseteq \dots$, it holds that the infinite sequence $s_0, s_1, \dots, s_n, \dots$ defined recursively as

$$\begin{aligned}s_0 &= C_0, \\ s_n &= s_{n-1} \Delta C_n\end{aligned}$$

is not strictly decreasing (and because the result of $C \Delta D$ is a lower bound of C , the sequence eventually stabilizes).

- Narrowing is performed only once a fixpoint is reached via widening.

Representation of a Program

- We choose (deterministic) **finite flowcharts** as a language independent representation of programs.
- A finite flowchart is a directed graph with 5 types of nodes:
 - entries,
 - assignments,
 - tests,
 - junctions,
 - exits.
- Abstract interpretation iteratively computes abstract contexts for each edge of the flowchart.
- An equation is associated with each edge of the flowchart according to the type of the tail node of the edge.

Representation of a Program

- **Entry**: denotes the entry point of a program. $C_O = \top$.

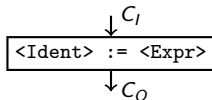


Representation of a Program

- **Entry**: denotes the entry point of a program. $C_O = \top$.



- **Assignment**: denotes the assignment A of expression $\langle \text{Expr} \rangle$ to the variable $\langle \text{Ident} \rangle$.
 $C_O = \tau(A, C_I)$.

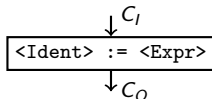


Representation of a Program

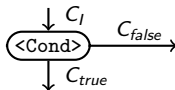
- **Entry:** denotes the entry point of a program. $C_O = \top$.



- **Assignment:** denotes the assignment A of expression $\langle \text{Expr} \rangle$ to the variable $\langle \text{Ident} \rangle$. $C_O = \tau(A, C_I)$.

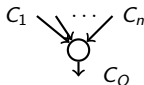


- **Test:** denotes splitting of the flow to branches B_{true} and B_{false} according to the Boolean condition $\langle \text{Cond} \rangle$. Two contexts are computed: $C_{true} = \tau(B_{true}, C_I)$ and $C_{false} = \tau(B_{false}, C_I)$.



Representation of a Program

- **Junction**: denotes join J of several branches of code execution (e.g., after `...then ...` and `...else ...` branches of an if statement or for a loop join).
 $C_O = \tau(J, C_1 \sqcup \dots \sqcup C_n)$.



It often holds for junctions that:

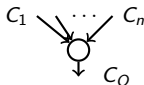
- $\tau(J) = \lambda x . x$ — for **simple junctions** (if branches),
- $\tau(J) = \lambda x . C_p \nabla x$ — for **loop junctions**,
- $\tau(J) = \lambda x . C_p \triangle x$ — for **loop junctions (only after widening)**

where C_p is the abstract context computed for the node in the previous iteration.

Representation of a Program

- **Junction**: denotes join J of several branches of code execution (e.g., after `...then ...` and `...else ...` branches of an if statement or for a loop join).

$$C_O = \tau(J, C_1 \sqcup \dots \sqcup C_n).$$



It often holds for junctions that:

- $\tau(J) = \lambda x . x$ — for **simple junctions** (if branches),
- $\tau(J) = \lambda x . C_p \nabla x$ — for **loop junctions**,
- $\tau(J) = \lambda x . C_p \triangle x$ — for **loop junctions (only after widening)**

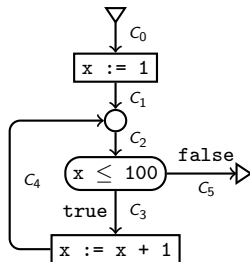
where C_p is the abstract context computed for the node in the previous iteration.

- **Exit**: denotes the exit point of a program.



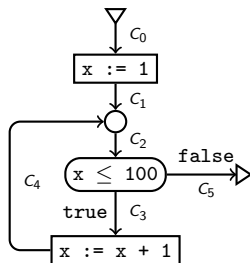
Program Example

- Consider the below flowchart program and analysis with the [interval abstract domain](#) $\mathbb{I}$:



Program Example

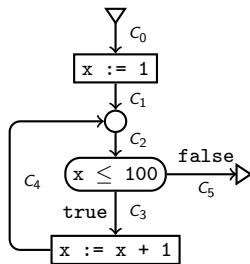
- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



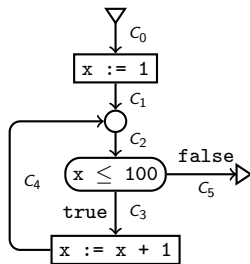
- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $\lfloor \cdot, \cdot \rfloor$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

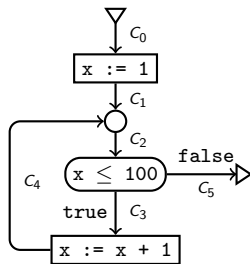
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$
- $C_2^0 = [,]$
- $C_3^0 = [,]$
- $C_4^0 = [,]$
- $C_5^0 = [,]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

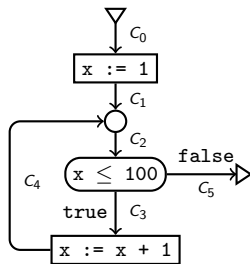
Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$
- $C_3^0 = [,]$
- $C_4^0 = [,]$
- $C_5^0 = [,]$

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

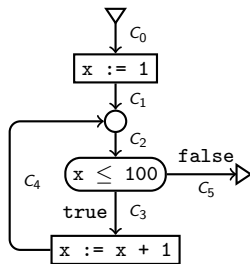
Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$
- $C_3^0 = [,]$
- $C_4^0 = [,]$
- $C_5^0 = [,]$

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

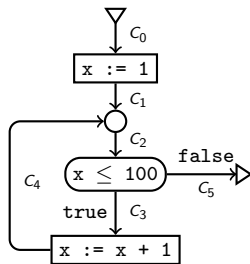
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$
- $C_4^0 = [,]$
- $C_5^0 = [,]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

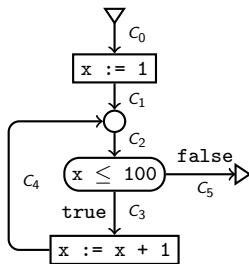
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$
- $C_4^0 = [,]$ $C_4^1 = [2, 2]$
- $C_5^0 = [,]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

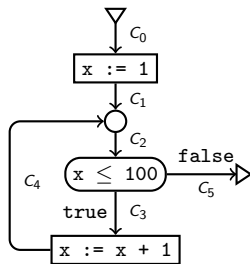
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$ $C_2^2 = [1, +\infty]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$
- $C_4^0 = [,]$ $C_4^1 = [2, 2]$
- $C_5^0 = [,]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

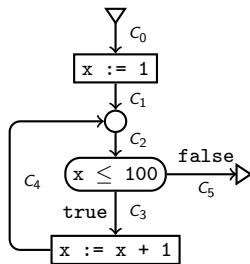
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$ $C_2^2 = [1, +\infty]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$ $C_3^2 = [1, 100]$
- $C_4^0 = [,]$ $C_4^1 = [2, 2]$
- $C_5^0 = [,]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

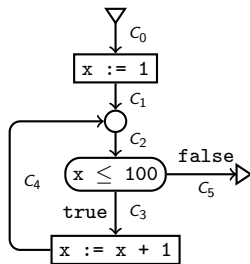
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$ $C_2^2 = [1, +\infty]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$ $C_3^2 = [1, 100]$
- $C_4^0 = [,]$ $C_4^1 = [2, 2]$ $C_4^2 = [2, 101]$
- $C_5^0 = [,]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- We will use notation $[a, b]$ for the predicate $a \leq x \leq b$ where $a, b \in \mathbb{Z} \cup \{-\infty, +\infty\}$.
- Assignments:** *interval arithmetic* (e.g., $[i, j] + [k, l] = [i + k, j + l]$).
- Join:** $[a, b] \sqcup [c, d] = [\min(a, c), \max(b, d)]$.
- Tests:** *intersections* of intervals.
- We define the **widening** ∇ of intervals as:
 - $[,]$ is the infimum of $(\mathbb{I}, \sqcup)$
 - $[i, j] \nabla [k, l] = [\text{if } k < i \text{ then } -\infty \text{ else } i, \text{if } l > j \text{ then } +\infty \text{ else } j]$.

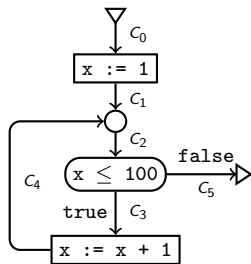
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$ $C_2^2 = [1, +\infty]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$ $C_3^2 = [1, 100]$
- $C_4^0 = [,]$ $C_4^1 = [2, 2]$ $C_4^2 = [2, 101]$
- $C_5^0 = [,]$ $C_5^1 = [101, +\infty]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \nabla (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- Let us now define the **narrowing** operation Δ over intervals as
 - $[i, j] \Delta [k, l] = [$
 if $i = -\infty$ then k else $\min(i, k)$,
 if $j = +\infty$ then l else $\max(j, l)$.
 $]$
- We now substitute the equation for C_2 with a new one that uses narrowing.

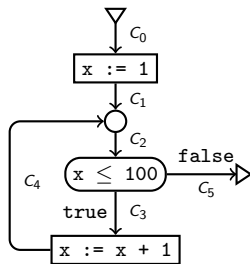
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$ $C_2^2 = [1, +\infty]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$ $C_3^2 = [1, 100]$
- $C_4^0 = [,]$ $C_4^1 = [2, 2]$ $C_4^2 = [2, 101]$
- $C_5^0 = [,]$ $C_5^1 = [101, +\infty]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \Delta (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- Let us now define the **narrowing** operation Δ over intervals as
 - $[i, j] \Delta [k, l] = [$
 if $i = -\infty$ then k else $\min(i, k)$,
 if $j = +\infty$ then l else $\max(j, l)$.
 $]$
- We now substitute the equation for C_2 with a new one that uses narrowing.

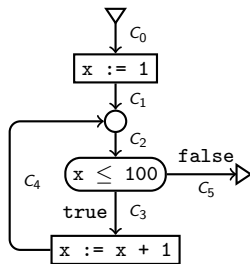
- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$ $C_2^2 = [1, +\infty]$ $C_2^3 = [1, 101]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$ $C_3^2 = [1, 100]$
- $C_4^0 = [,]$ $C_4^1 = [2, 2]$ $C_4^2 = [2, 101]$
- $C_5^0 = [,]$ $C_5^1 = [101, +\infty]$

Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \Delta (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

Program Example

- Consider the below flowchart program and analysis with the **interval abstract domain** $\mathbb{I}$:



- Let us now define the **narrowing** operation Δ over intervals as
 - $[i, j] \Delta [k, l] = [$
 if $i = -\infty$ then k else $\min(i, k)$,
 if $j = +\infty$ then l else $\max(j, l)$.
 $]$
- We now substitute the equation for C_2 with a new one that uses narrowing.

- $C_0^0 = [-\infty, +\infty]$
- $C_1^0 = [,]$ $C_1^1 = [1, 1]$
- $C_2^0 = [,]$ $C_2^1 = [1, 1]$ $C_2^2 = [1, +\infty]$ $C_2^3 = [1, 101]$
- $C_3^0 = [,]$ $C_3^1 = [1, 1]$ $C_3^2 = [1, 100]$
- $C_4^0 = [,]$ $C_4^1 = [2, 2]$ $C_4^2 = [2, 101]$
- $C_5^0 = [,]$ $C_5^1 = [101, +\infty]$ $C_5^2 = [101, 101]$

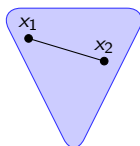
Equations that describe the **flow of the analysis**; their fix-point solution represents results of the analysis:

- $C_0 = [-\infty, +\infty]$
- $C_1 = [1, 1]$
- $C_2 = C_2 \Delta (C_1 \sqcup C_4)$
- $C_3 = C_2 \cap [-\infty, 100]$
- $C_4 = C_3 + [1, 1]$
- $C_5 = C_2 \cap [101, +\infty]$

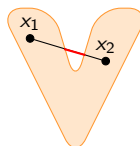
Polyhedral Analysis

Polyhedral Analysis

- An abstract interpretation-based approach of automatic discovery of relations among program variables expressible as **linear inequations**,
 - this can be seen as a generalization of the interval analysis.
- Let $\vec{x} = x_1, \dots, x_n \in \mathbb{R}^n$ be the variables of a program. We can use a **convex polyhedron** $P \subseteq \mathbb{R}^n$ to represent a set of assignments to $\vec{x}$.
- We use **convex** polyhedra because operations on them are reasonably efficient (a set $C \subseteq \mathbb{R}^n$ is convex iff $\forall x_1, x_2 \in C, \forall 0 \leq \lambda \leq 1 : \lambda x_1 + (1 - \lambda)x_2 \in C$).



convex set



non-convex set

- **Use:** compile-time determination of bounds of variables, discovery of constants, ...

Representation of a Convex Polyhedron

- We use two dual ways to represent a convex polyhedron:
 - by a **system of linear inequations**, and
 - by the **frame of the polyhedron**.
- We can alter between these representations (with some overhead).
- Efficient execution of different operations require different representation.

System of Linear Inequations

- Let $\vec{x} = x_1, \dots, x_n \in \mathbb{R}^n$ be the variables of a program. Given a finite set of m linear inequations over $\vec{x}$ of the form

$$\left\{ \sum_{i=1}^n a_{ji} x_i \leq b_j \mid 1 \leq j \leq m \right\}$$

or equivalently using vectors and matrices as

$$\vec{x} \cdot \mathbf{A} \leq \vec{b},$$

we can geometrically interpret the solutions of the inequations as a **convex polyhedron** in $\mathbb{R}^n$ defined by the intersection of *halfspaces* corresponding to each inequality.

The Frame of a Convex Polyhedron

- A convex polyhedron P can also be characterized by its **frame** $F = (V, R, L)$:
 - **Vertices** V : points $\vec{v}$ of a polyhedron P that are not *convex combinations* of other points $\{\vec{w}_1, \dots, \vec{w}_m\}$ of P ,

$$\left(\left(\vec{v} = \sum_{i=1}^m \lambda_i \vec{w}_i \right) \wedge (\forall 1 \leq i \leq m : (\vec{w}_i \in P \wedge \lambda_i \geq 0)) \wedge \left(\sum_{i=1}^m \lambda_i = 1 \right) \right) \Rightarrow \\ \Rightarrow (\forall 1 \leq i \leq m : (\lambda_i = 0 \vee \vec{w}_i = \vec{v})).$$

- **Convex hull**: the set of all convex combinations of V .

The Frame of a Convex Polyhedron

- **Extreme rays** R : rays $\vec{r}$ of P (i.e. vectors such that there exists a half-line parallel to $\vec{r}$ and entirely included in P) that are not positive combinations of other rays $\vec{s}_1, \dots, \vec{s}_p$ of P :

$$\left(\vec{r} = \sum_{i=1}^p \mu_i \vec{s}_i \wedge (\forall 1 \leq i \leq p : \mu_i \in \mathbb{R}^+) \right) \Rightarrow (\forall 1 \leq i \leq p : (\mu_i = 0 \vee \vec{s}_i = \vec{r})).$$

The Frame of a Convex Polyhedron

- **Extreme rays** R : rays $\vec{r}$ of P (i.e. vectors such that there exists a half-line parallel to $\vec{r}$ and entirely included in P) that are not positive combinations of other rays $\vec{s}_1, \dots, \vec{s}_p$ of P :

$$\left(\vec{r} = \sum_{i=1}^p \mu_i \vec{s}_i \wedge (\forall 1 \leq i \leq p : \mu_i \in \mathbb{R}^+) \right) \Rightarrow (\forall 1 \leq i \leq p : (\mu_i = 0 \vee \vec{s}_i = \vec{r})).$$

- **Lines** L : vectors $\vec{l}$ such that both $\vec{l}$ and $-\vec{l}$ are rays of P :

$$\forall \vec{x} \in P, \forall \mu \in \mathbb{R} : \vec{x} + \mu \vec{l} \in P.$$

The Frame of a Convex Polyhedron

- **Extreme rays** R : rays $\vec{r}$ of P (i.e. vectors such that there exists a half-line parallel to $\vec{r}$ and entirely included in P) that are not positive combinations of other rays $\vec{s}_1, \dots, \vec{s}_p$ of P :

$$\left(\vec{r} = \sum_{i=1}^p \mu_i \vec{s}_i \wedge (\forall 1 \leq i \leq p : \mu_i \in \mathbb{R}^+) \right) \Rightarrow (\forall 1 \leq i \leq p : (\mu_i = 0 \vee \vec{s}_i = \vec{r})).$$

- **Lines** L : vectors $\vec{l}$ such that both $\vec{l}$ and $-\vec{l}$ are rays of P :

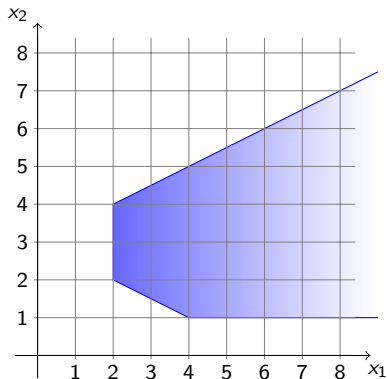
$$\forall \vec{x} \in P, \forall \mu \in \mathbb{R} : \vec{x} + \mu \vec{l} \in P.$$

- Every point $\vec{x}$ of the polyhedron P defined by the frame $F = (V, R, L)$ can be obtained from V , R and L :

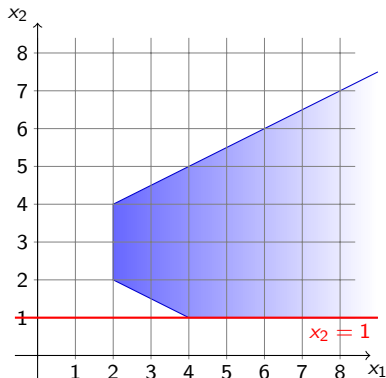
$$\vec{x} = \sum_{i=1}^{\sigma} \lambda_i \vec{v}_i + \sum_{j=1}^{\rho} \mu_j \vec{r}_j + \sum_{k=1}^{\delta} \nu_k \vec{l}_k$$

where $0 \leq \lambda_1, \dots, \lambda_{\sigma} \leq 1, \sum_{i=1}^{\sigma} \lambda_i = 1, \mu_1, \dots, \mu_{\rho} \in \mathbb{R}^+, \nu_1, \dots, \nu_{\delta} \in \mathbb{R}.$

Example of a Convex Polyhedron



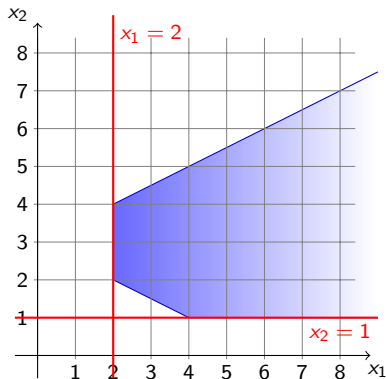
Example of a Convex Polyhedron



System of linear inequations

$$x_2 \geq 1$$

Example of a Convex Polyhedron

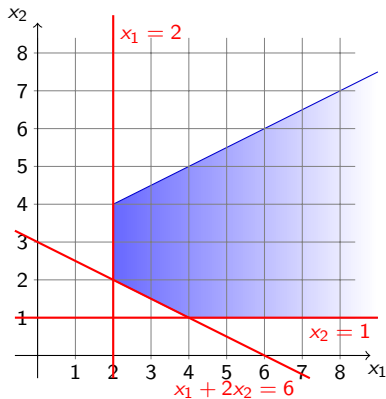


System of linear inequations

$$x_2 \geq 1$$

$$x_1 \geq 2$$

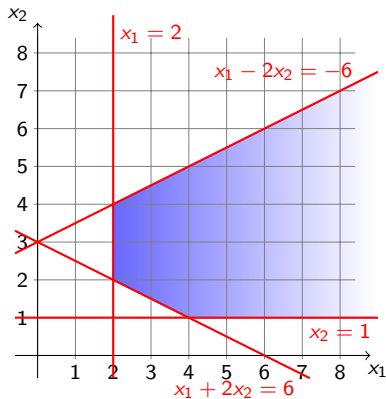
Example of a Convex Polyhedron



System of linear inequations

$$\begin{aligned}x_2 &\geq 1 \\x_1 &\geq 2 \\x_1 + 2x_2 &\geq 6\end{aligned}$$

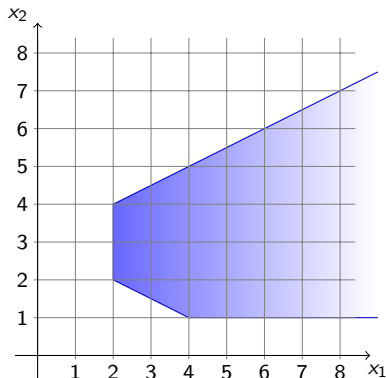
Example of a Convex Polyhedron



System of linear inequations

$$\begin{aligned}x_2 &\geq 1 \\x_1 &\geq 2 \\x_1 + 2x_2 &\leq 6 \\x_1 - 2x_2 &\geq -6\end{aligned}$$

Example of a Convex Polyhedron



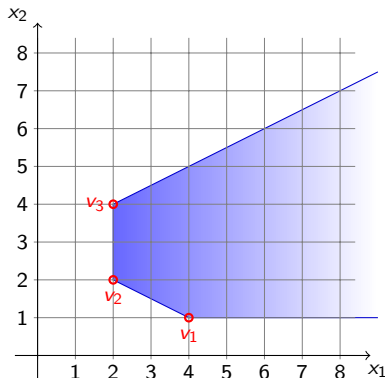
System of linear inequations

$$\begin{aligned}x_2 &\geq 1 \\x_1 &\geq 2 \\x_1 + 2x_2 &\geq 6 \\x_1 - 2x_2 &\geq -6\end{aligned}$$

Frame of a polyhedron

$$F = (V, R, L)$$

Example of a Convex Polyhedron



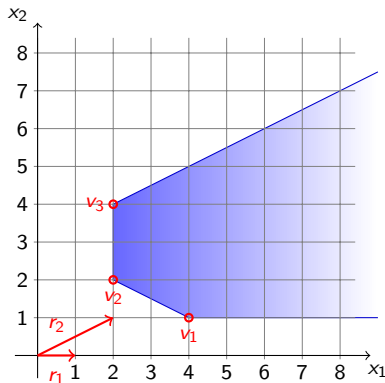
System of linear inequations

$$\begin{aligned}x_2 &\geq 1 \\x_1 &\geq 2 \\x_1 + 2x_2 &\geq 6 \\x_1 - 2x_2 &\geq -6\end{aligned}$$

Frame of a polyhedron

$$\begin{aligned}F &= (V, R, L) \\V &= \{\vec{v}_1 = [4, 1], \vec{v}_2 = [2, 2], \vec{v}_3 = [2, 4]\}\end{aligned}$$

Example of a Convex Polyhedron



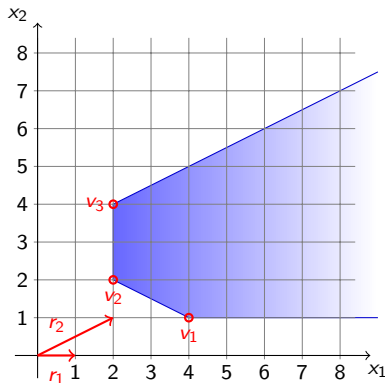
System of linear inequations

$$\begin{aligned}x_2 &\geq 1 \\x_1 &\geq 2 \\x_1 + 2x_2 &\geq 6 \\x_1 - 2x_2 &\geq -6\end{aligned}$$

Frame of a polyhedron

$$\begin{aligned}F &= (V, R, L) \\V &= \{\vec{v}_1 = [4, 1], \vec{v}_2 = [2, 2], \vec{v}_3 = [2, 4]\} \\R &= \{\vec{r}_1 = (1, 0), \vec{r}_2 = (2, 1)\}\end{aligned}$$

Example of a Convex Polyhedron



System of linear inequations

$$\begin{aligned}x_2 &\geq 1 \\x_1 &\geq 2 \\x_1 + 2x_2 &\geq 6 \\x_1 - 2x_2 &\geq -6\end{aligned}$$

Frame of a polyhedron

$$\begin{aligned}F &= (V, R, L) \\V &= \{\vec{v}_1 = [4, 1], \vec{v}_2 = [2, 2], \vec{v}_3 = [2, 4]\} \\R &= \{\vec{r}_1 = (1, 0), \vec{r}_2 = (2, 1)\} \\L &= \emptyset\end{aligned}$$

Transformations of Convex Polyhedra

- Different types of nodes of the flowchart representation of a program perform distinct transformation on the polyhedron. The number of input and output polyhedra differs according to the type of the node.
- **Entries:** create a polyhedron according to constraints on the input values of variables (in case there are none for variable x_i , the polyhedron is unbounded in i -th dimension).

Assignments

- Performed operations vary according to assigned expression:
 - **Non-linear expression** $x_i := \langle \text{non-linear expression} \rangle$: because these cannot be represented using convex polyhedra, any constraint on x_i is dropped (we add line $\vec{d}$ to frame such that $d_i = 1$ and $\forall 1 \leq j \leq n, i \neq j : d_j = 0$).
 - **Linear expression** $x_i := \sum_{j=1}^n a_j x_j + b$: the frame $F' = (V', R', L')$ of the output polyhedron can be computed from the frame $F = (V, R, L)$ of the input as

Assignments

- Performed operations vary according to assigned expression:
 - **Non-linear expression** $x_i := \langle \text{non-linear expression} \rangle$: because these cannot be represented using convex polyhedra, any constraint on x_i is dropped (we add line $\vec{d}$ to frame such that $d_i = 1$ and $\forall 1 \leq j \leq n, i \neq j : d_j = 0$).
 - **Linear expression** $x_i := \sum_{j=1}^n a_j x_j + b$: the frame $F' = (V', R', L')$ of the output polyhedron can be computed from the frame $F = (V, R, L)$ of the input as
 - $V' = \{\vec{v}'_1, \dots, \vec{v}'_\sigma\}$ where $\vec{v}'_j$ is defined by $v'_{ji} = \vec{a} \vec{v}_j + b$ and $v'_{jm} = v_{jm}$ where $\forall 1 \leq m \leq \sigma, m \neq i$.

Assignments

- Performed operations vary according to assigned expression:
 - **Non-linear expression** $x_i := \langle \text{non-linear expression} \rangle$: because these cannot be represented using convex polyhedra, any constraint on x_i is dropped (we add line $\vec{d}$ to frame such that $d_i = 1$ and $\forall 1 \leq j \leq n, i \neq j : d_j = 0$).
 - **Linear expression** $x_i := \sum_{j=1}^n a_j x_j + b$: the frame $F' = (V', R', L')$ of the output polyhedron can be computed from the frame $F = (V, R, L)$ of the input as
 - $V' = \{\vec{v}'_1, \dots, \vec{v}'_\sigma\}$ where $\vec{v}'_j$ is defined by $v'_{ji} = \vec{a} \vec{v}_j + b$ and $v'_{jm} = v_{jm}$ where $\forall 1 \leq m \leq \sigma, m \neq i$.
 - $R' = \{\vec{r}'_1, \dots, \vec{r}'_\rho\}$ where $\vec{r}'_j$ is defined by $r'_{ji} = \vec{a} \vec{r}_j$ and $r'_{jm} = r_{jm}$ for $\forall 1 \leq m \leq \rho, m \neq i$.

Assignments

- Performed operations vary according to assigned expression:
 - **Non-linear expression** $x_i := \langle \text{non-linear expression} \rangle$: because these cannot be represented using convex polyhedra, any constraint on x_i is dropped (we add line $\vec{d}$ to frame such that $d_i = 1$ and $\forall 1 \leq j \leq n, i \neq j : d_j = 0$).
 - **Linear expression** $x_i := \sum_{j=1}^n a_j x_j + b$: the frame $F' = (V', R', L')$ of the output polyhedron can be computed from the frame $F = (V, R, L)$ of the input as
 - $V' = \{\vec{v}'_1, \dots, \vec{v}'_\sigma\}$ where $\vec{v}'_j$ is defined by $v'_{ji} = \vec{a} \vec{v}_j + b$ and $v'_{jm} = v_{jm}$ where $\forall 1 \leq m \leq \sigma, m \neq i$.
 - $R' = \{\vec{r}'_1, \dots, \vec{r}'_\rho\}$ where $\vec{r}'_j$ is defined by $r'_{ji} = \vec{a} \vec{r}_j$ and $r'_{jm} = r_{jm}$ for $\forall 1 \leq m \leq \rho, m \neq i$.
 - $L' = \{\vec{l}'_1, \dots, \vec{l}'_\delta\}$ where $\vec{l}'_j$ is defined by $l'_{ji} = \vec{a} \vec{l}_j$ and $l'_{jm} = l_{jm}$ for $\forall 1 \leq m \leq \delta, m \neq i$.

Tests

- The input polyhedron P is transformed into two output polyhedra: P_t for the true branch and P_f for the false branch.
- For a Boolean condition C , it needs to hold that $P_t \supseteq P \cap T_C$, $P_f \supseteq P \setminus T_C$ where T_C is the subset of $\mathbb{R}^n$ such that each point of T_C satisfies C (the right sides of the inclusions are not necessarily convex polyhedra).
- The operation that is performed varies according to the Boolean condition of the test:

Tests

- The input polyhedron P is transformed into two output polyhedra: P_t for the true branch and P_f for the false branch.
- For a Boolean condition C , it needs to hold that $P_t \supseteq P \cap T_C$, $P_f \supseteq P \setminus T_C$ where T_C is the subset of $\mathbb{R}^n$ such that each point of T_C satisfies C (the right sides of the inclusions are not necessarily convex polyhedra).
- The operation that is performed varies according to the Boolean condition of the test:
 - **Non-linear tests:** $P_t = P_f = P$ (can be refined for some cases).

Tests

- The input polyhedron P is transformed into two output polyhedra: P_t for the true branch and P_f for the false branch.
- For a Boolean condition C , it needs to hold that $P_t \supseteq P \cap T_C, P_f \supseteq P \setminus T_C$ where T_C is the subset of $\mathbb{R}^n$ such that each point of T_C satisfies C (the right sides of the inclusions are not necessarily convex polyhedra).
- The operation that is performed varies according to the Boolean condition of the test:
 - **Non-linear tests:** $P_t = P_f = P$ (can be refined for some cases).
 - **Linear equality tests:** Boolean condition $C : \vec{a}\vec{x} = b$ defines a *hyperplane* H . If P is included in H , then $P_t = P, P_f = \emptyset$. If P is not included in H , then $P_t = P \cap H$ and $P_f = P$.

Tests

- The input polyhedron P is transformed into two output polyhedra: P_t for the true branch and P_f for the false branch.
- For a Boolean condition C , it needs to hold that $P_t \supseteq P \cap T_C$, $P_f \supseteq P \setminus T_C$ where T_C is the subset of $\mathbb{R}^n$ such that each point of T_C satisfies C (the right sides of the inclusions are not necessarily convex polyhedra).
- The operation that is performed varies according to the Boolean condition of the test:
 - **Non-linear tests:** $P_t = P_f = P$ (can be refined for some cases).
 - **Linear equality tests:** Boolean condition $C : \vec{a}\vec{x} = b$ defines a *hyperplane* H . If P is included in H , then $P_t = P, P_f = \emptyset$. If P is not included in H , then $P_t = P \cap H$ and $P_f = P$.
 - **Linear inequality tests:** for Boolean condition $\vec{a}\vec{x} \leq b$, the outputs are $P_t = P \cap \vec{a}\vec{x} \leq b$ and $P_f = P \cap \vec{a}\vec{x} \geq b$.

Junctions

- Junctions correspond to merge of several program paths so the output polyhedron P is union of all input polyhedra P_i . It is computed according to the kind of the junction:

Junctions

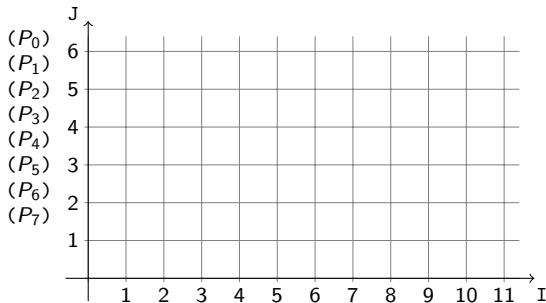
- Junctions correspond to merge of several program paths so the output polyhedron P is union of all input polyhedra P_i . It is computed according to the kind of the junction:
 - **Simple junctions:** for input polyhedra $P_1, \dots, P_m$, we compute the convex hull of $P_1 \cup \dots \cup P_m$.

Junctions

- Junctions correspond to merge of several program paths so the output polyhedron P is union of all input polyhedra P_i . It is computed according to the kind of the junction:
 - **Simple junctions:** for input polyhedra $P_1, \dots, P_m$, we compute the convex hull of $P_1 \cup \dots \cup P_m$.
 - **Loop junctions:** for input polyhedra $P_1, \dots, P_m$, let Q be the convex hull of $P_1 \cup \dots \cup P_m$. Then $P' = P \nabla Q$ is the convex polyhedron consisting of linear constraints of P satisfied by every element of Q .

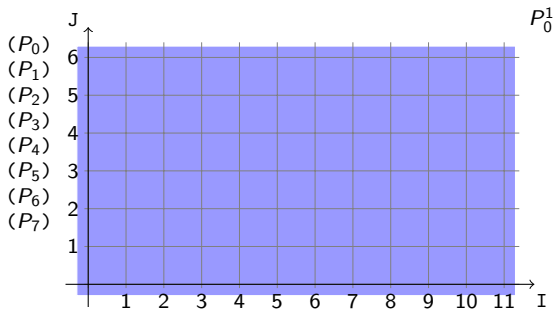
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
go to L;
```



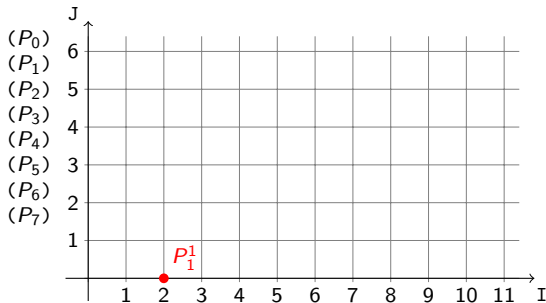
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
go to L;
```



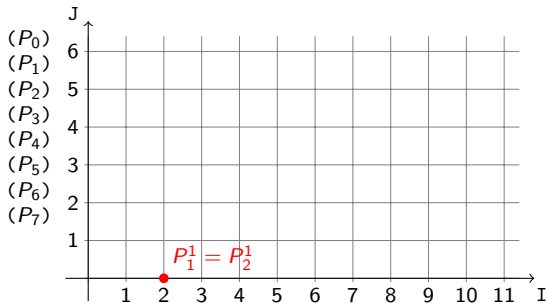
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



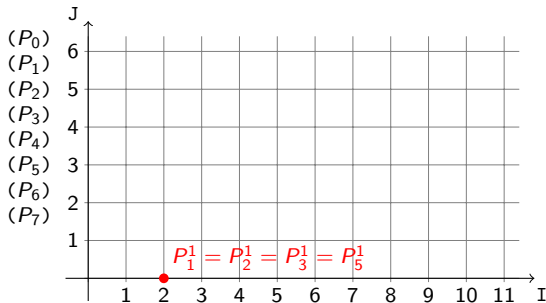
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
go to L;
```



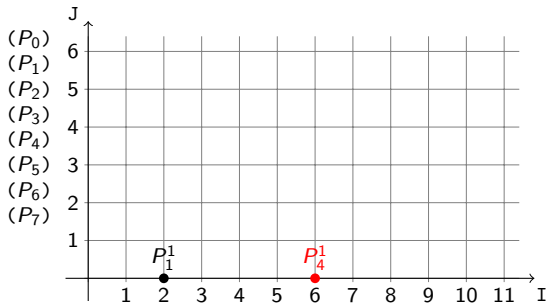
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



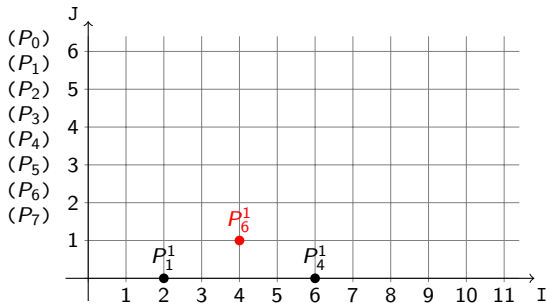
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



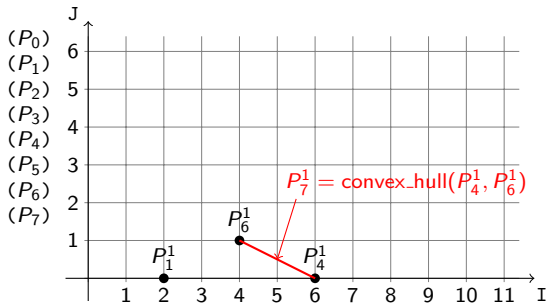
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



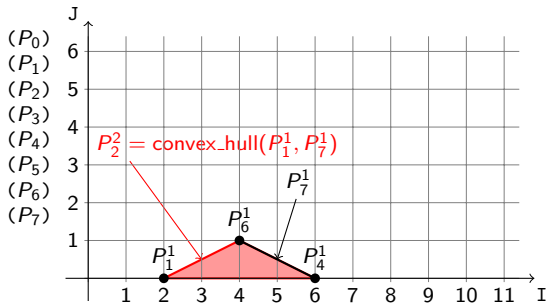
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



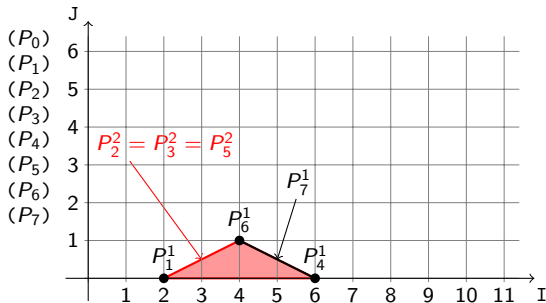
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



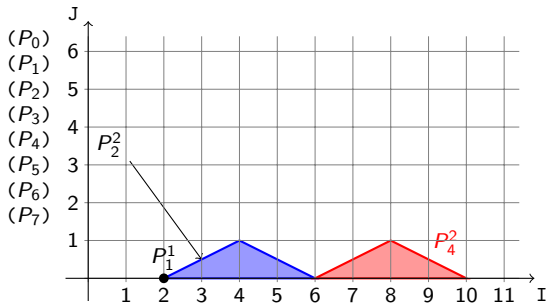
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



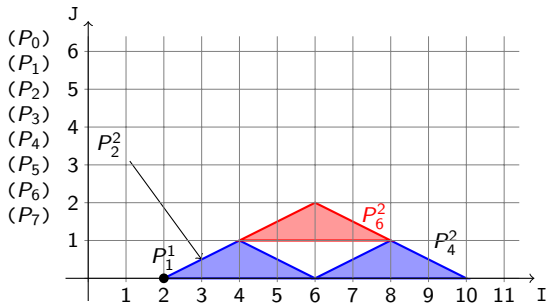
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



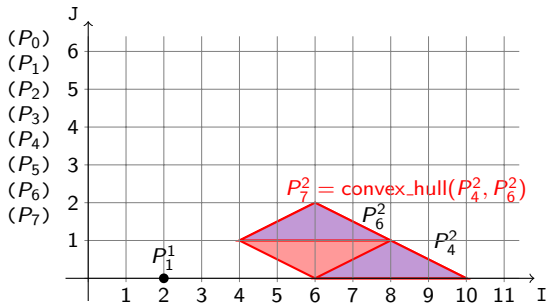
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



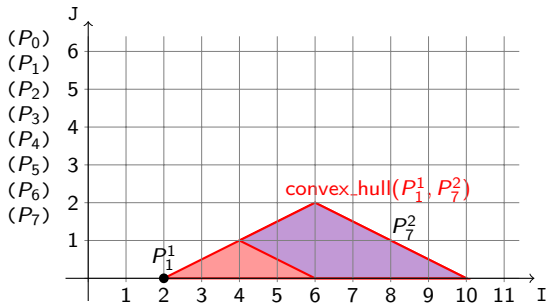
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



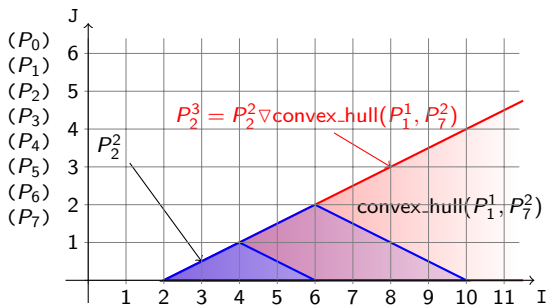
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



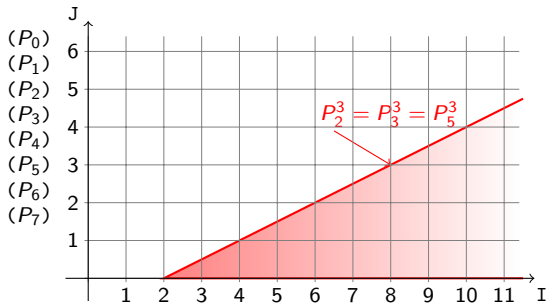
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



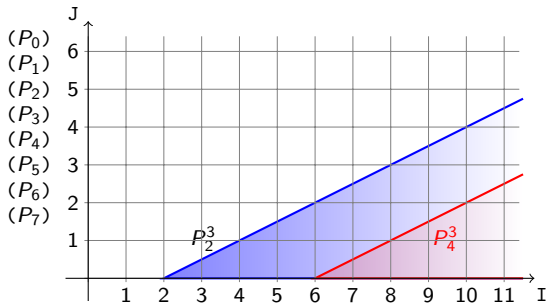
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



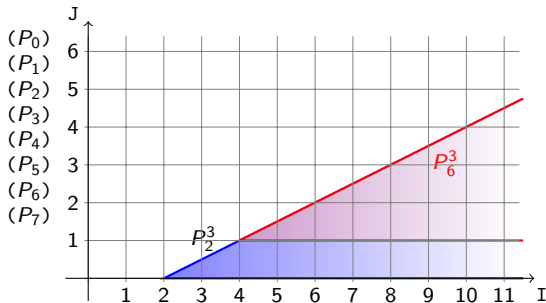
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



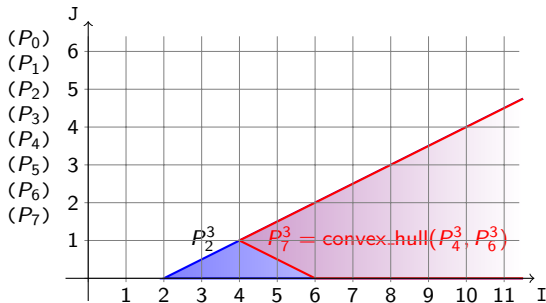
Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



Example

```
I := 2, J := 0;  
L:  
  if ... then  
    I := I + 4  
  else  
    J := J + 1, I := I + 2;  
  fi;  
  go to L;
```



Various Known Domains

Numerical Domains

Typically, **conjunctions of constraints** of some form for $a, b, c \in \mathbb{Z} \cup \{-\infty, +\infty\}$:

- **Signs**: $x [\leq | < | = | > | \geq] 0, x = [\perp | \top]$.
- **Intervals**: $a \leq x \leq b$.
- **Difference Bound Matrices**: $x - y \leq c, [+|-]x \leq c$.
- **Octagons**: $[+|-]x [+|-]y \leq c, [+|-]x \leq c$.
- **Polyhedra**: $\vec{a}\vec{x} \leq b$.
- **Simple Congruences**: $x = a \bmod b$.
- **Linear Congruences**: $\vec{a}\vec{x} = b \bmod c$.
- ...

Facebook Infer bounds analysis – for buffer sizes and values of indices, intervals with the following kinds of bounds are used:

$[+|-]\infty$, linear expressions, $a[+|-][\min|\max](b, x)$.

Improving the Precision

The precision of numerical domains can be improved by various kinds of **partitioning**:

- Characterizing abstract contexts reachable **at each program line separately** (used already in the example with interval analysis).
- Tracking values up to some preconfigured bound **precisely**.
- Cutting concrete domains of some variables into some preconfigured sets of **regions** (e.g., $(-\infty, a_1)$, $[a_1, a_2)$, $[a_2, a_3)$, ..., $[a_n, +\infty)$ for some $n > 1$ and $a_i < a_{i+1}$ for $1 \leq i \leq n - 1$) and then **widening within the regions only**.

Shape Analysis

- **Shape analysis**: analysis of **shapes of dynamic data structures** (not just a may/must point-to analysis).
- Quite complex: **characterising infinite sets of graphs!**
- Many different domains: one of them is separation logic.
- A fragment of **separation logic with a list predicate**:
 - **Pure part** – Boolean combinations of $x = [y|NULL]$.
 - **Shape predicates**:
 - $x \mapsto [y|NULL]$ – **points-to** predicate.
 - $ls(x, [y|NULL])$ – non-empty, singly-linked **list** predicate.
 - Shape predicates joined by the **separating conjunction** “*”.
Conjoined heaps must allocate different memory cells, e.g.:
 - $x \mapsto y * x \mapsto z$ is **false**,
 - while $x \mapsto y * ls(z, y) * u \mapsto v$ is **satisfiable**.

Shape Analysis – Continued

- The pure and spatial parts are conjoined.
- The variables are either **program variables** or **existentially quantified logical variables** (denoting anonymous memory cells).
- Separation allows for **modular analysis**.
- Computing **sets of formulae** per location (a **top-level disjunction**).
- **Abstract transformers** for pointer manipulating statements based on (partial) **concretisation**:
 - $ls(x, y) \equiv x \mapsto y \vee \exists z : x \mapsto z * ls(z, y)$.
 - Then manipulate $x \mapsto y$.
- **Join**: **union of sets of formulae**, perhaps with **pruning** those covered by other (weaker) formulae (e.g., $x \mapsto y \sqsubseteq ls(x, y)$).
- **Widening**: based on **abstracting** sequences of points-to and list segments through logical variables to a single list segment:
 - $\alpha(\exists y, z : x \mapsto y * ls(y, z) * z \mapsto NULL) \equiv ls(x, NULL)$.

Abstract Regular Model Checking

- **Regular model checking** (RMC): system configurations $\sim$ words, sets of configurations $\sim$ regular languages (finite automata).
- **Transitions**: finite transducers or special operations on automata.
- **Abstract RMC** – can be viewed as abstract interpretation with widening that abstracts automata by collapsing some states:
 - Overapproximation wrt language inclusion.
 - **States to collapse**: same languages of words up to some bound or intersecting the same predicate languages.
 - Abstraction is automatically refineable using a CEGAR loop.
- **Generalisations**: trees and nested forests with leaf-to-root references – (tuples of (nested)) tree automata.
- **Applications**: parametric protocols (mutual exclusion, ...), recursion, communication queues, strings, heap structures (lists, trees, ...), microprocessor pipelining, ...

Interprocedural Analysis

Summaries

- **Summaries**: (sets of) pre-/post-condition pairs for functions.
- Record under which **precondition** a function can be executed leading to a given **postcondition**.
- Pre-/post-conditions encoded using abstract contexts recording **relevant parts of the entire contexts**: values of parameters, relevant global variables, accessible parts of heap, ..., returned/changed parts of contexts.
- Can be **computed top-down**:
 - If a function has **never been called** with the precondition, analyse it and record a **new pair** into its summary.
 - If a function has **already been called** under the same precondition, **use directly the recorded result**.
 - If the input context is **covered** ($\sqsubseteq$) by some recorded precondition: may use the postcondition for a **loss of precision**.

Summaries: Bottom-Up

- Summaries can also be computed **bottom-up** along the **call-tree**.
- Each function analysed **without any known context**: needs to derive under which precondition(s) a function can produce post-condition(s) meaningful for the given analysis.
- For example:
 - If analysing pointer safety and seeing a dereference without a check, derive that the pointer should be non-null.
 - If analysing locking and seeing a lock without a preceding unlock, derive that the pointer should be unlocked.
- Can be **very scalable**: each function analysed precisely once!
- Easier to **lose information**.
- **Harder to design** the analysis.